

Analysis of Sailplane Flight Logs to Evaluate In-Flight Engine Start Procedures

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Abstract—This project analyzes thousands of IGC flight log files to enhance sailplane safety by documenting in-flight engine start procedures and evaluating pilot adherence to manufacturer guidelines. In-flight engine starts allow pilots to return to base after atmospheric convective activity ends, yet little statistical analysis exists on compliance with recommended procedures. By identifying patterns in real-world flight data, detecting anomalies, and assessing pilot decision-making, this project highlights current practices and their implications for safety. The insights gained will support aviation safety organizations in developing data-driven recommendations for safer sailplane operations.

Keywords—*IGC flight logs, in-flight engine start, false positive detection, pilot behavior, sailplane safety*

I. INTRODUCTION

Sailplane pilots may need to perform in-flight engine starts to return to base, especially in instances when atmospheric conditions are no longer sufficient for gliding. However, challenges during engine starts can pose safety risks and may result in off-field landings. This project analyzes IGC flight logs to assess pilot adherence to manufacturer-recommended engine start procedures, identify anomalies, and document current practices for safer operations. By understanding pilot behavior and identifying trends related to engine start attempts, the team aimed to highlight current practices and enhance safety protocols.

The project uses IGC flight logs which are ASCII text files containing both mandatory and optional variables recorded at regular time intervals. Some of the mandatory variables include timestamp, latitude, longitude, GPS altitude, pressure altitude, and engine noise level (ENL). Optional variables can include airspeed. An important assumption of this project is that the IGC flight log data accurately captures these variables. The project focuses on analyzing IGC logs, identifying trends, and classifying likely engine start attempts. It does not include real-time monitoring or diagnostic solutions.

The findings from this project may support pilots, glider manufacturers, and aviation safety organizations by providing data-driven insights into operational practices.

II. Project Goals

- A. to report on adherence to safety procedures
- B. to analyze the in – flight operation of the engine to determine if an engine start was successful or not.
- C. What engine noise thresholds yield the fewest number of false positives?

III. LITERATURE REVIEW AND EXISTING WORK

A. Literature Review

In terms of the existing literature on IGC file analysis specifically, there was a master’s thesis called “Geoinformatics in Gliding” [2] that worked on an algorithm that maximizes distance flown during free flight glider competitions. Distance flown in this type of competition is the sum of distances flown between multiple points connected sequentially, not just the straight line from takeoff to landing. By carefully choosing turning points, pilots can maximize the distance flown during the competition. Without turn points, the distance measured would simply be the direct straight line from takeoff to landing, usually much shorter. The core computational challenge is finding the combination of these fixes as turn points that yield the greatest total cumulative distance. This isn't trivial, as the number of possible combinations grows rapidly. In the study, the team used dynamic programming to accomplish the goal. The paper was also concerned with establishing an open data exchange mechanism for real-time glider competition data. They were trying to find a standardized way of recording flight data, and one of their key recommendations is to engineer a device that can populate the same format of IGC file, which is developed by the world authority of gliding, FAI-IGC. The envisioned standard focuses on capturing data for any given glider competition, not just a single glider.

B. EXISTING WORK

There are numerous libraries in python that can be leveraged to help analyze IGC files. One of those is `igc-processor`. While this package is not installable with `pip`, the team looked at the Github repo and used some of the functions in the repo such as “`is_circling`” to detect circling segments during the flights. Another library that is commonly used amongst people trying to analyze files is `libigc`. It helps you identify circling segments during flights but in a slightly different way than `igc-processor`. In the future, it would be interesting to compare the two methods of these libraries to detect circling segments and compare accuracy to known circling segments. Furthermore, `libigc` is installable with `pip` and is a fantastic resource because it has easy-to-use methods. Future work could consider the data engineering process to use methods such as “`vertical_velocity`”. We have an altitude rate of change during the 5-minute window before each engine start event, but not vertical velocity.

IV. DATA DESCRIPTION

Within the IGC file, each line item has a specific purpose. In every IGC file, there is one I-record. If a line starts with I, it is the I-record, and its structure shows how the B-record will be set up. In the I-record, the 3 letter sequences represent the different data points that the flight recorder is picking up. The preceding 4 numbers are the positions in the B-record where that data is found. A B-record is an ASCII text string with mandatory and optional variables. Mandatory variables are time, lat and long, engine noise, pressure, and GPS altitude. Optional variables depend on the sensors installed and the logger’s manufacturer design. If the line starts with B, it is a B-record.

All the information within each B-record must have a data issue time within 0.1 seconds of the time given in the B-record. The following table shows an example of the B-record (FAI & IGC 2001).

B	HHMMSS	DDMMMM	N/S	DDMMMM	E/W	V/A	PPPPP	GGGGG	CR	LF
B	114412	4617200	N	02001758	E	A	00827	00864		

The given example means that at 11:44:12 UTC, at 46°12.200' N, and at 20°01.758' E. In case of V/A using A for a 3D fix and V for a 2D fix (no GPS altitude) or for no GPS data. Altitude above 1013.25 hPa sea level datum is 827, the altitude above the WGS84 ellipsoid is 864 (FAI & IGC 2001). This capstone project mainly uses the information from the B-record.

V. METHODOLOGY

Our method of obtaining insights from these B-records was to create a data frame of relevant statistics for each engine start event within each flight. We focused on the five minutes preceding each engine start, as we recognized that this period may reveal insights about the pilot’s thought process. We

calculated statistics such as the terrain elevation during the engine run, the height above ground level at the start of the engine, average true air speed during the engine run, and more to gain insight into pilot intentions regarding the in-flight operation of the engine.

Our first step was to filter out flights without any engine start events. Once we created a script that could automate this process, we were left with about 2,000 flights out of the original 30,000 flights we received. Next, we used geospatial engineering to create relevant features for our IGC logs.

To answer our first problem statement, we produced descriptive analytics based on which glider manufacturers and glider types intended to start their engine below 1,000ft. To obtain the right numbers for this analysis, we had to understand our three different types of gliders. We have pure gliders, self-launchers, and turbo gliders. Pure gliders were excluded from the analysis, because those don’t have engines. Self-launchers have an engine that is used in the beginning of the flight to get the plane off the ground, so this first engine start event was not accounted for in the 1,000ft analysis. Self-launchers will always start their engine below 1,000ft, so we looked at only the second engine start events and onward for self-launchers. We evaluated all turbo glider engine starts. This was because engine starts of turbo gliders are supposed to be started above 1,000ft according to safety protocols. A flight has an optimal speed to start the engine, known as the blue line speed. We flagged flights that were out of bounds of the expected blue line speed, which we said was 50 to 65 knots. If the pilot was going much faster than 65 knots, it could damage the plane and is unsafe. If the pilot was going too far below 50 knots, then the engine will not operate.

The most critical variables that provide insights on whether pilots are following safety protocols are average speed during the engine run and height above ground level (AGL) that the engine started. Most log files did not have true air speed (TAS) indications in the B-records. Therefore, we needed to use python functions to identify circling segments to get the wind vector. We subtracted the wind vector during circling segments (assumed to be the same during the engine run) from the instantaneous ground speed vector at each timestamp during the engine run. Instantaneous ground speed was calculated as the difference in distance between the two positions at each timestamp divided by the time difference between those two consecutive points. To verify that these calculations were reasonable, we compared the calculated value of TAS with the recorded value.

Our team was unable to address the second problem statement of whether engine starts were successful because flight loggers do not record the necessary data points needed to answer this question. One method to capture this data is for the manufacturers to provide data for when the engine was

engaged. If this data is provided, we can then investigate instances where the engine was engaged but the noise threshold was not crossed for longer than 60 seconds. This would tell us that the pilot tried to start the engine, but it failed. However, we were successful in describing pilot behavior as it relates to height loss during engine start events. We identified numerous features that were highly correlated with height loss during each engine start, which has a strong impact on glider safety.

A false positive engine start was one in which the average speed during the engine start was below 50 knots and one with height loss. Once we removed these false positives, we created numerous plots showing which features were most strongly associated with height loss. Height loss is not expected during an engine start, as the entire objective of starting the engine is to gain height. The variable height-gain loss is the net altitude change from the start of the engine start to the end.

We also created two data sets, one with the original thresholds, and one with increased thresholds. We evaluated if there were fewer flights that started their engine below 1,000ft with the increased thresholds and those plots can be seen below.

III. RESULTS AND DISCUSSION

After cleaning our data, we were able to get a more precise picture of real engine start events. First, we removed any events where the engine was started within the first 5 minutes of the flight, since those were likely to be test runs. That left us with 709 out of 873 events to analyze, which is about 81% of the original data. Next, we removed cases where the glider lost height during the engine start, which brought the number down to 487 events, or 55%. This was removed since the glider is supposed to gain height at an engine start event, not lose height. Suggesting that it was a false positive. Finally, we removed events where the average true air speed (TAS) was less than 50 knots. After that, we had 417 events left to analyze, which is about 48% of the original data. After filtering out false positives and test engine starts, we were left with 417 meaningful engine start events out of our original 2,000 flights. Our goal was to understand what conditions led to successful engine start events—defined here as those that resulted in height gain—and how those conditions varied by altitude, glider type, terrain, and other influencing factors. It also should be noted that in our analysis we focused on glider types to isolate pilot behavior.

We found a strong relationship between altitude above ground level (AGL) at the time of engine start and the likelihood of height gain. The density plot of engine start AGL by height gain indicator showed a clear divide between cases of height loss and height gain. Engine starts that failed to produce height gain tended to occur around or below 1,000 ft AGL, peaking close to this threshold and dropping off quickly after 2,000 ft. In contrast, engine start events with height gain clustered higher, peaking around 3,500 ft AGL and extending past 5,000 ft. There was a minimal overlap between these two distributions, reinforcing the validity of the 1,000 ft AGL guideline. This

visualization supports the claim that pilots who initiate engine starts at higher altitudes greatly improve their chances of achieving a positive climb.

When looking at glider types, the ASW 27-18 E (ASG-29E) glider had the highest number of engine start events below 1000 ft AGL—over 60 events. However, percentage-wise, the AS 33ES stood out even more, with around 70% of its engine start events below the threshold. This suggests that some glider types (or perhaps the pilots who tend to fly them) are more likely to engage the engine earlier than recommended. This could be due to perceived aircraft capabilities, pilot confidence, or training, but it raises concerns since the data clearly shows lower altitude starts correlate with poorer outcomes. In our top ten gliders that had engine start events below 1,000 ft, we tend to see that while pilots follow standards at times, they skew towards starting their engines too early.

Another factor that played a part in whether the pilot would experience height gain during their engine start was the distance traveled during the preceding 5-minute window. We found that having a greater distance traveled during the preceding 5-minute window, about 5-10 miles, as compared to traveling 1-5 miles is associated with height gain. Covering 5–10 statute miles in the five-minute pre-start window equates to an average groundspeed of roughly 60–120 kt—well into a brisk cruise regime for most turbos and self-sustainers. Pilots who are still travelling that far have typically remained in a comparatively *energy-rich* flight state. By contrast, flights that cover only 1–5 miles ($\approx$ 12–60 kt) are already close to min-sink or even in a steepening descent/circling pattern; the lower kinetic reserve leaves less time for the engine to spool up before the glider's energy is spent, increasing the odds that the run produces little or no net height gain.

Terrain also played a notable role in engine start decision-making. Our terrain label distribution plot revealed that engine starts associated with height loss were more likely to take place over open forests, herbaceous vegetation, and sparse vegetation. Meanwhile, height gain during engine starts was more common over agriculture, closed forests and evergreen needleleaf. One possible interpretation is that pilots delay starting their engines until they are over terrain perceived as safer in the event of a failed start—like open fields or farmland. However, this strategy also pushes the engine start closer to the ground, where the probability of a successful climb is significantly reduced. These findings suggest that pilots may be prioritizing terrain safety at the expense of altitude, which could be counterproductive given that height is one of the most critical safety factors. To provide a more quantitative rather than categorical view of terrain, we calculated the terrain elevation at the time of the engine start and compared these values with height gain and height loss. As expected, a lower terrain elevation at the time of the engine start is associated with height gain.

The team also examined pilot air speed during engine start events. We filtered out engine events where the calculated average true air speed fell below 50 knots, since those are outside the safe operating envelope. The resulting subset showed that maintaining proper air speed is another indicator of success. By calculating true air speed from ground speed adjusted for wind (derived from circling segments), we validated that most successful engine start events occurred within a higher speed range (about 60-80 knots range) than the original blue line speed provided by manufacturers. This insight flags a topic for dialogue with manufacturers - whether the published blue-line is not high enough to allow most pilots to obtain height gain while the engine is running.

Our speed calculations were 5% off for turbo gliders and 7.5% off for self-launch gliders in terms of overall accuracy as compared to known values in 118 flights. Without close speed approximations for the remaining flights, we would not have had a representative enough sample size to draw conclusions about pilot safety and pilot behavior.

Although we initially set out to detect failed engine starts—cases where the pilot likely attempted an engine start but it didn't engage—the lack of binary indicators in the IGC logs (such as whether the button was pressed to engage engine) made it difficult to isolate these events. focused on proxy indicators such as altitude loss and insufficient speed during engine start events, labeling these as false positives. This gave us a more realistic view of the dataset by filtering out misleading records.

IV. CONCLUSION

Our first problem statement was to determine whether the community is adhering to the manufacturer's guidelines for in-flight engine operations. We investigated whether the guideline was followed on starting the engine at 1,000 ft above ground level (AGL). We performed analysis looking at which gliders had the highest number of engine start events below 1,000 ft AGL. Through the analysis, we tend to see that while most pilots follow the manufacturer's standard of starting the engine at 1,000 ft AGL, we do tend to see that pilots skew towards starting the engine too early. When investigating further on these successful engine start outcomes where pilots did follow the manufacturer's standard, we show a clear relationship with engine start altitude, terrain, and air speed. Pilots who start the engine above 1,000 ft AGL and maintain appropriate air speed have a significantly higher chance of achieving climb. Glider type and terrain further influence these outcomes, with some gliders consistently showing higher rates of below-threshold

starts. These insights highlight areas for improved pilot training and awareness.

Our second problem statement was to establish a method to determine whether an in-flight engine start attempt was successful or failed. A failed engine start is defined as instances where the pilot intentionally attempted to start the engine, but it failed to engage. We were unable to address this due to the data availability. We recommend the manufacturers install necessary sensors to provide data for when an engine is engaged. The technology is available to provide binary data on whether certain buttons were pressed or engaged. This technology should be utilized by the manufacturers and provided to analysts for future work. However, we did provide analysis that helped describe pilot behavior as it relates to which variables were correlated to height loss during engine start events, which impacts glider safety.

Lastly, for our third problem, we compared two datasets with regular noise threshold and increased noise threshold (increased by 100 for each noise indicator) to see how the ratio changes for how many slights there were that had engine starts below a 1,000 ft AGL. This was an effort to understand how the thresholds might impact false positives. We learned that:

1. Higher thresholds do not mean that more pilots successfully start above 1,000 ft AGL
2. There are relatively equal proportion of engine starts above and below 1,000 ft for each set of thresholds
3. Higher noise level overall during the engine run period is not associated with more flights having height gain.

References

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V. PLOTS

Figure 1

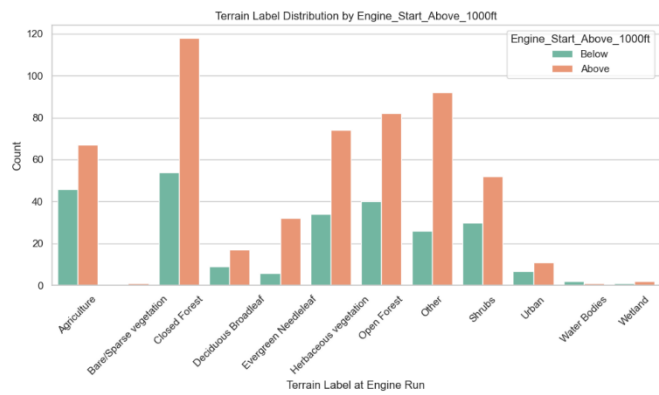


Figure 2

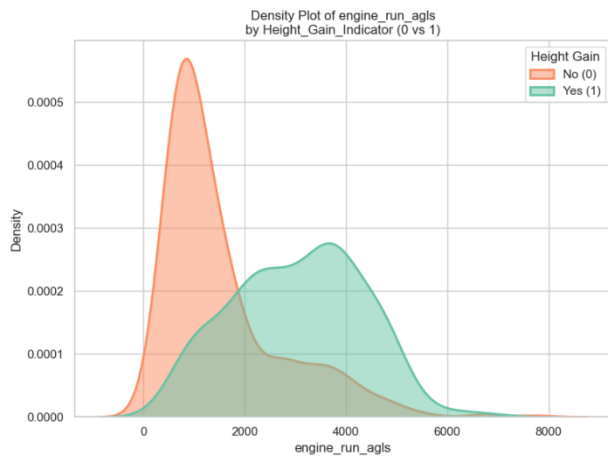


Figure 3

